VERANDA

Find. Verify. Settle.

DAY 3 · PRODUCT ARCHITECTURE, SOFTWARE INTEGRATION & MLOPS

Refined System Architecture

Target architecture, failure register and a lean MLOps plan for Veranda: one view of every Lagos rental listing in your areas, plus a verified report before you pay.

PRESENTED BY

Abayomi Oshinowo

Founder, Veranda

SLIDE 1

Target architecture

4 layers · 16 components · 5 services

SLIDE 2

Failure register

3 failures · thresholds · rollbacks

SLIDE 3

Tooling & 2-week backlog

lean tools · 9 tasks · testable done

November 2026 · yourveranda.com · 23 Lagos areas · Built on the Audos platform

Refined target architecture

Every interface, store and monitor labelled. Orange tags mark the refinements from lab feedback and the Nov 2026 scraper review.

NEW refinement - - - server-side trust boundary

1 DATA INGESTION

Task scheduler
Daily crons (Lagos): NPC 06:00, PPro 06:45
photo backfills ×3 · commute baselines ×3

Listing scraper (Apify) NEW
NPC cheerio · PPro firefox headful, NG proxy
per-batch insert + dedupe, resume after kill

Lookup collectors
OpenCellID → area_coverage
Google Routes v2 → area/listing/address routes

Official + crowd data
LASEMA/NiMet, NERC bands, tariffs → area_profiles
tenant submissions → tenant_reports

2 MODEL & SCORING SERVING

Rules scoring engine
Good / Fair / Watch out / Improving badges
commute <30 · 30–60 · >60 min; security unscored

Grounded report Q&A (LLM) NEW
OpenAI proxy; answers only from this report
allow-listed model + golden eval set

Price normaliser NEW
annual rent only; amount + period parsed apart
rejects counted: period/amount/range/other

Customer agent
Claude Agent SDK (platform-hosted)
grounded in the customer prompt

3 API GATEWAY / BACKEND

WorkspaceDB data API
14 tables; DB token ships in the page
trusted writes only via server functions

Server functions (4)
scrape · coverage · commutes · FX drift
600 s run ceiling, ~300 s per outbound fetch

Secrets & Apify proxies
BYOK Google + OpenCellID keys stay server-side
Apify runs proxied and billed by platform

Auth & payments
email one-time-code sessions
Stripe NGN 7,000/mo ($5.04), hosted checkout

4 FRONTEND

Find & Verify app
Verify · Browse · Report · Account
5-card report + Before You Sign debt check

"Ask about this report"
chat under every unlocked report
says "not measured yet" instead of guessing

Admin dashboard
password-gated: listings, accounts, MRR
tenant reports: verify / edit / delete

In-space agent + Pitch Deck
app-first landing; assistant one tap away
free sample report for prospects

SUPPORTING SERVICES · DATABASES, TELEMETRY, ALERT PATHS

Databases
listings (10.7k) · area_profiles (23) ·
tenant_reports · accounts & unlocks

Telemetry: scrape_runs NEW
one row per run; status ok / degraded / partial / failed / interrupted, per-URL counts

Lease: scrape_locks NEW
primary-key lease stops overlapping runs; stale leases taken over

Versioning
workspace snapshots + server-function code fingerprints; baseline kept for rollback

Alert path NEW
degraded / failed run → email founder ·
weekly FX drift >2% → email (live)

Security boundaries
Browser holds only the DB token + session. API keys live in the secrets proxy. Admin is password-gated; card data stays in Stripe Checkout.

Fallbacks (never invent a number)
Gray "Improving" / "Data unavailable" · commute: exact route → area-to-hub baseline → "temporarily unavailable + Retry" · coverage: published-report baseline.

Human review
Tenant reports get the "verified" tick only after founder review. Hook updates are registered only after dry-run, simulated-interrupt and SQL checks. Price rebases are founder-approved.

Top 3 failure modes, thresholds & rollbacks

Each failure is already visible in production data. Every signal has a number, every rollback an owner and a recovery time.

FAILURE MODE	EVIDENCE TODAY	SIGNAL & ALERT THRESHOLD	ROLLBACK / RECOVERY STEPS	RECOVERY TIME	OWNER
1. Silent ingestion failure (data drift) Listing data stops or degrades while runs report success.	• NPC: 0 items on every run since 30 Sep (40+ runs logged ok=true) • PropertyPro runs 519/520 killed mid-run, all rows lost	• status ≠ ok on any run → email same day • 0 items from all URLs → degraded (ok=false) • inserted = 0 on 2 consecutive days → incident • run open > 11 min → interrupted	1. Re-register previous hook by fingerprint 2. Next run resumes unfinished batches (no duplicates) 3. NPC blocked → move to browser actor + residential proxy 4. Run dedupe-cleanup dry run; confirm 0 duplicate IDs	**Detect ≤ 24 h** **Rollback ≤ 15 min** **Data back next run**	**Eng (Otto / build agent)**
2. Lookup provider outage (latency / errors) Commute and network cards cannot get live measurements.	• 92/92 commute baselines + 16 routes pending • 92/92 coverage rows pending (keys refused)	• pending > 20% of a matrix 48 h after schedule • ≥ 3 consecutive provider errors (PERMISSION_DENIED / unknown_secret) • route call p95 > 10 s	1. Cards fall back instantly: baseline / "Improving" / Retry 2. Fix key restrictions (Routes API, OpenCellID) 3. Pause the 3 daily baseline schedules until fixed 4. Re-run measure-areas; check pending share	**Fallback: instant** **Fix ≤ 1 day** **Matrix full ≤ 48 h**	**Founder (keys) + Eng**
3. Report Q&A hallucination / model break Chat states facts not in the report, or stops answering.	• Chat pinned to gpt-4o-mini; proxy only accepts allow-listed models • No grounding eval set yet	• chat error rate > 5% of asks per day • any golden-set answer with a number not in context • ≥ 1 renter flag per week	1. Kill switch: hide chat card (report still works) 2. Revert to last good prompt version 3. Switch to allow-listed model (gpt-6.1-sol, medium) 4. Re-run 30-question golden set before re-enabling	**Kill switch ≤ 15 min** **Model swap ≤ 1 h**	**Eng; founder signs off**

Alert path for all three: scrape_runs / provider status → daily health query → email to founder (platform.sendEmail). Recovery times are targets to be confirmed by the drills in the Slide 3 backlog.

Lean MLOps plan & 2-week backlog

Tools a one-founder team can actually run, and a sprint ordered by risk reduced per unit of effort. Sprint: Mon 16 – Fri 27 Nov 2026.

LEAN TOOL SELECTION

Versioning

Workspace snapshots · server-function code fingerprints · baseline source kept beside each draft

Not: separate Git/CI platform

Testing

node:test offline harness (45 tests, mocked DB + Apify, sanitised HTML fixtures) · 30-question chat golden set

Not: paid test cloud

CI / CD

Gated manual release: dry run → simulated interrupt → SQL checks → register; rollback = re-register baseline

Not: Kubernetes, Airflow

Observability

scrape_runs run log + daily health SQL + founder email alerts; FX drift monitor already live

Not: Datadog, MLflow

PRIORITISED BACKLOG · RISK REDUCTION vs EFFORT

#	TASK	RISK	EFFORT	OWNER	DEFINITION OF DONE (TESTABLE)
1	**Register reviewed scraper update + create scrape_locks**	F1	S	Eng	Live fingerprint 12379f54…; dry run + simulated interrupt pass; 0 duplicate external_ids
2	**Unblock Google Routes key (add Routes API to key)**	F2	S	Founder	≥ 1 area_commutes row measured; pending < 20% after 48 h
3	**Unblock OpenCellID key**	F2	S	Founder	≥ 1 area_coverage row measured; card shows "live lookup" source
4	**Daily health check → founder email**	F1-3	S	Eng	Forced degraded test run sends email within 15 min
5	**Chat to allow-listed model + kill switch**	F3	S	Eng	Error rate < 5% on 50 test asks; toggle hides card in < 15 min
6	**Move NPC scraping to browser actor + residential proxy**	F1	M	Eng	NPC status ok with ≥ 300 items on 3 consecutive days
7	**30-question grounding golden set for chat**	F3	M	Eng	≥ 95% pass; 0 invented numbers; set saved with the tests
8	**Review 5 suspected merged-digit rents**	F1	S	Founder	Each row corrected or confirmed; rule recorded in run summary
9	**Rollback drill + one-page runbook**	F1-3	S	Eng	Each failure restored in a drill inside its recovery-time target

Shaded rows 1–5 = week 1 (highest risk reduced per effort), rows 6–9 = week 2. Risk = failure mode reduced (Slide 2). Effort: S ≤ 1 day, M 2–3 days.

How this blueprint meets the rubric

Strong vs weak criteria from the Day 3 blueprint rubric, and where each is answered.

	STRONG LOOKS LIKE	✓ HOW VERANDA ANSWERS IT
Completeness	**All interfaces, DBs, monitors labelled**	Slide 1 labels 4 layers, 16 components, 5 supporting services, the trust boundary, fallbacks and human review. Each arrow is a real interface (Apify, proxies, DB API, OTP, Stripe).
Proportional tools	**Lean tools match team skills**	Snapshots, fingerprints, node:test, a run-log table and email alerts. All are already on the platform and run by one founder. No Kubernetes, Airflow, MLflow or Datadog.
Actionability	**Thresholds, owners, tested steps**	Slide 2 gives a numeric threshold, numbered rollback, recovery target and owner per failure. Slide 3 gives every task an owner, effort and a testable definition of done.